

Optimization for Deep Learning

SGD, Momentum, Adam, and Learning-Rate Schedules

Prof. Nipun Batra

ES 667 · Deep Learning · IIT Gandhinagar

From backprop to optimization

Where we are

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How do we turn a stream of gradients into good parameters? That is the job of the **optimizer**.

The spine of this lecture

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optimization decides how to use them.

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each step: current params → gradient → update rule → new params

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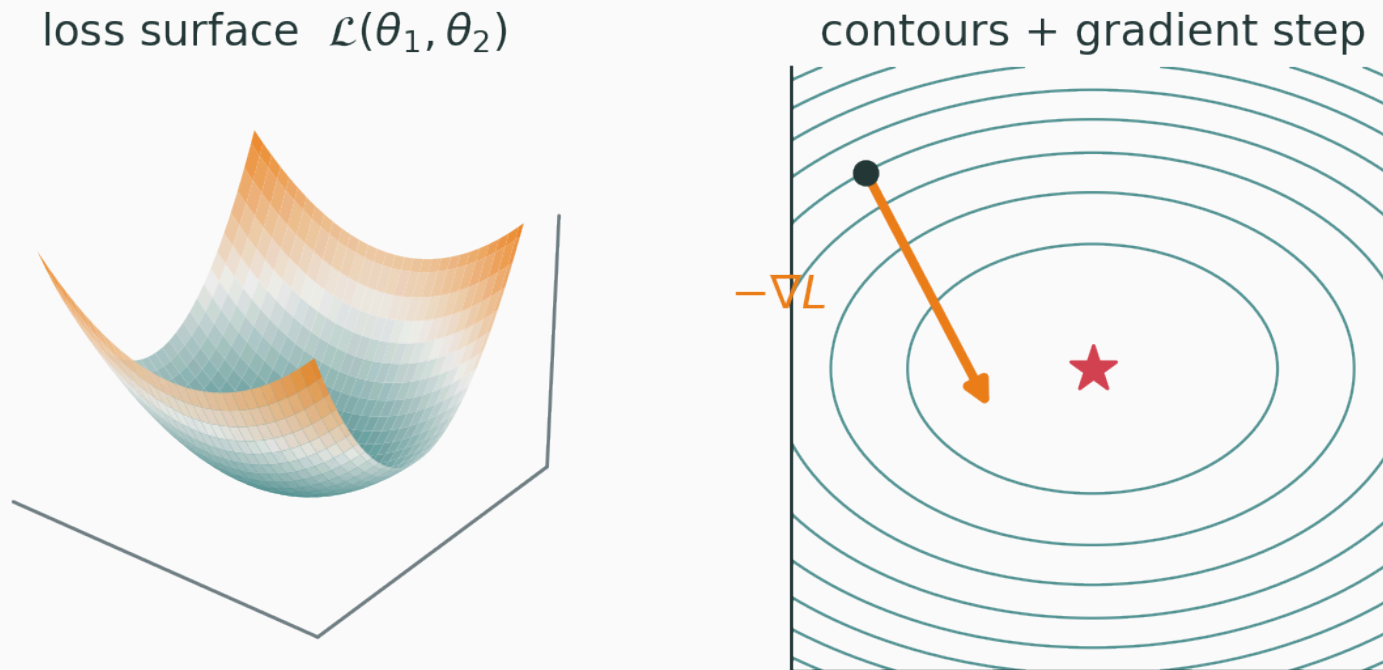
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- plain GD keeps **nothing** — just $-\eta g_t$;
- momentum keeps a **running direction**;
- Adam keeps a running **direction** and a running **magnitude**.

Training = descending a surface $\mathcal{L}(\theta)$ over millions of parameters.

The loss landscape

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the gradient points **uphill**; we step along $-\nabla \mathcal{L}$, perpendicular to contours

Expand the loss around the current point θ for a small step Δ :

$$\mathcal{L}(\theta + \Delta) \approx \mathcal{L}(\theta) + \nabla \mathcal{L}(\theta)^\top \Delta + \frac{1}{2} \Delta^\top H \Delta$$

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Keep only the **first-order** term (small steps $\Rightarrow$ curvature negligible):

$$\mathcal{L}(\theta + \Delta) \approx \mathcal{L}(\theta) + \nabla \mathcal{L}(\theta)^\top \Delta$$

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η plays the role of an **inverse-curvature** / trust-region size. A scalar η is a crude stand-in for H^{-1} — too big overshoots sharp directions, too small crawls on flat ones.

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- η — the **learning rate** (step size);
- $\nabla \mathcal{L}$ — the direction of steepest ascent, so $-\nabla \mathcal{L}$ descends;
- one hyperparameter, no memory — the simplest optimizer there is.

Worked example: 1-D quadratic

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the gradient -6 points left of the min, so the step moves us **right**, toward 3.

Worked example: 1-D quadratic (cont.)

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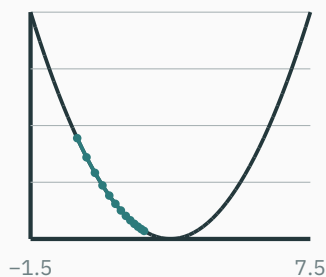
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The distance to the minimum obeys $(\theta_t - 3) \rightarrow (1 - 2\eta)(\theta_t - 3)$; here $1 - 2\eta = 0.8$, a clean geometric decay.

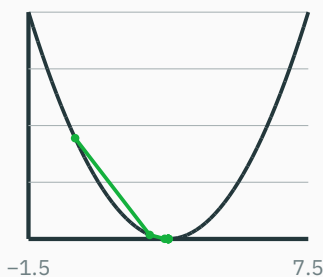
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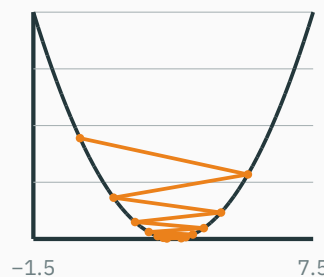
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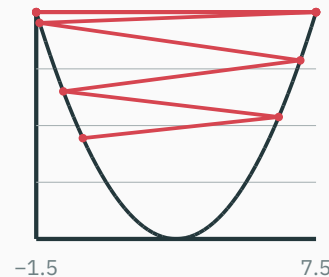
$\eta = 0.4$ · good



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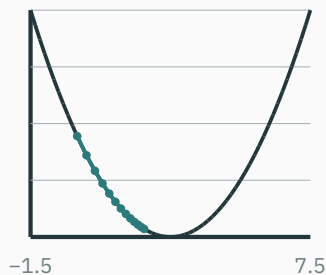
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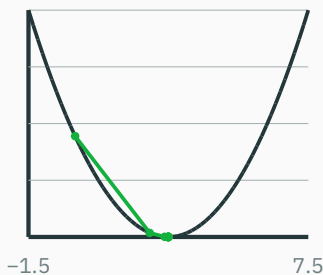
The learning rate matters

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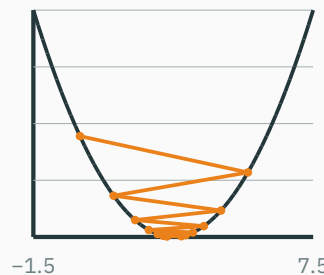
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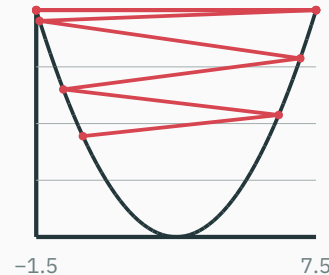
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too small crawls · good converges fast · large oscillates · $\eta > 1$ diverges

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/optimizer-race

Drop a start point on a loss surface and watch gradient descent trace its path as you sweep the learning rate — see crawl, converge, oscillate, and diverge live.

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Try values around the stability boundary and watch the error multiplier $1 - 2\eta$.

For $\mathcal{L} = (\theta - 3)^2$, gradient descent diverges for which learning rates?

A. $\eta < 0$ only

B. $0 < \eta < 0.5$

C. $0.5 < \eta < 1$

D. $\eta > 1$

Correct: D — $\eta > 1$

Why: The error is multiplied by $1 - 2\eta$ each step. Convergence requires its magnitude to be below one; beyond $\eta = 1$ it grows.

Full-batch vs stochastic gradient descent

The loss is an average

Training loss is an **empirical risk** — a mean over n examples:

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell_i(\theta), \quad \nabla \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \nabla \ell_i(\theta)$$

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The true gradient needs a pass over the **entire dataset**. For $n = 10^6$ that is a million backprops per step.

Full-batch gradient descent

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{1}{n} \sum_{i=1}^n \nabla \ell_i(\theta_t)$$

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Pro — the exact gradient; smooth, deterministic descent.

Con — one step costs a **full epoch**; far too slow for large n .

Stochastic gradient descent (SGD)

Sample one example $I \sim \text{Uniform}(1, \dots, n)$ and step on **its** gradient:

$$g_t = \nabla \ell_I(\theta_t), \quad \theta_{t+1} = \theta_t - \eta g_t$$

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one example per step — up to $n \times$ more updates per epoch

Take the expectation over the random index I :

$$\mathbb{E}_I[g_t] = \sum_{i=1}^n \Pr(I = i) \nabla \ell_i(\theta_t) = \sum_{i=1}^n \frac{1}{n} \nabla \ell_i(\theta_t)$$

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on average, SGD steps in the **true gradient direction**

Unbiased, but noisy

Write the stochastic gradient as the truth plus zero-mean noise:

$$g_t = \nabla \mathcal{L}(\theta_t) + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) > 0$$

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Each step is right **on average** but wrong **individually**. The path jitters — that noise both hurts (never fully settles) and helps (escapes bad regions).

Minibatch SGD — the practical middle

Average the gradient over a **batch** B_t of B examples:

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larger batch $\Rightarrow \sqrt{B}$ times less gradient noise

The batch-size tradeoff

Batch size	Gradient	Behaviour
small (1–32)	very noisy	cheap steps, explores, can escape saddles
medium (64–512)	moderate	the usual sweet spot
large (4k+)	near-exact	stable but costly; needs LR tuning

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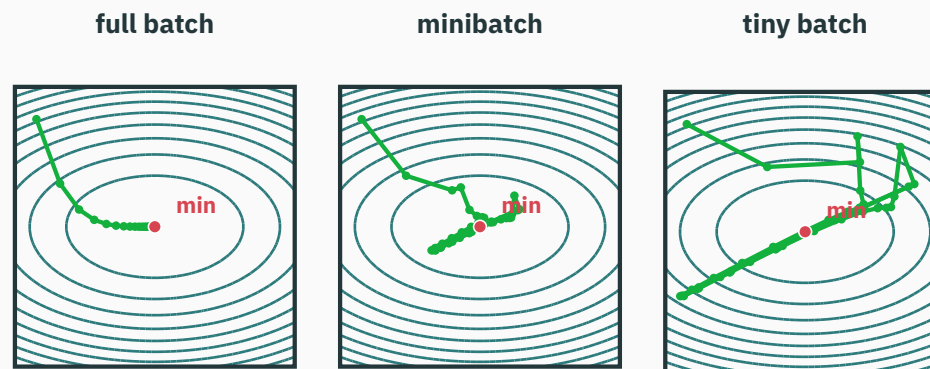
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Batch size is set as much by **GPU memory and throughput** as by optimization.

Same bowl, three noise levels (full / mini / tiny batch):

Batch size shapes the noise

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full batch glides · minibatch jitters toward the minimum · tiny batch bounces

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Switch between full-batch and stochastic updates and watch the trajectory go from a smooth glide to a noisy walk as the batch size shrinks.

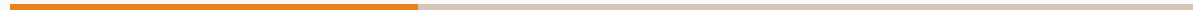
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Compare a smooth full-batch path with the small random deviations of minibatches.

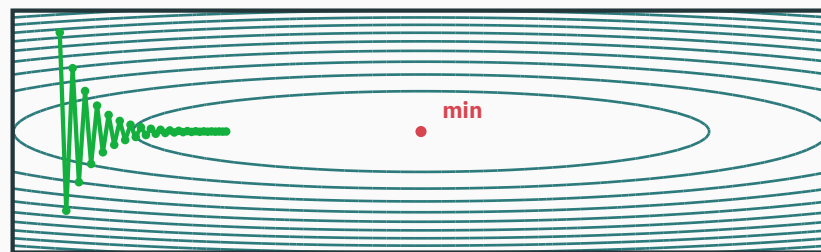
Momentum



Ill-conditioned loss $\mathcal{L} = x^2 + 100y^2$: steep across, shallow along.

The problem: ravines

Ill-conditioned loss $\mathcal{L} = x^2 + 100y^2$: steep across, shallow along.



GD must use a **tiny** η to stay stable in y — so it **zigzags** and crawls in x

The idea: keep a velocity

Accumulate past gradients into a **velocity**, then step along it:

$$v_t = \beta v_{t-1} + g_t, \quad \theta_{t+1} = \theta_t - \eta v_t$$

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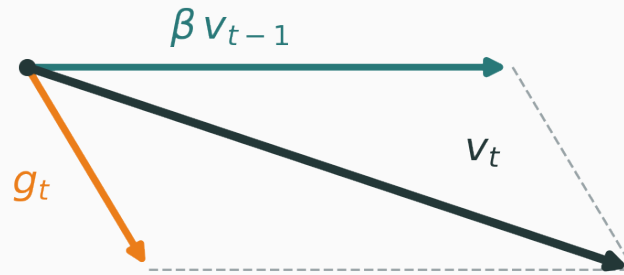
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- $\beta \in [0, 1)$ — the **momentum** coefficient (typically 0.9);
- v_t is a running memory of **where we have been going**.

The update is a **vector addition**: the decayed velocity plus this step's gradient.

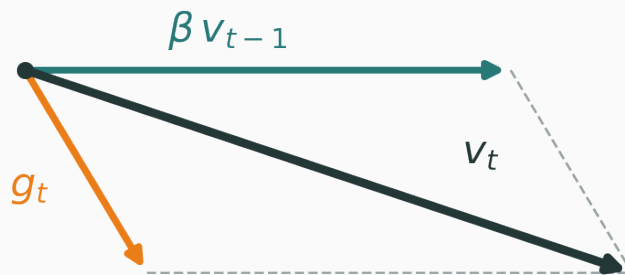
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the long βv_{t-1} dominates; the cross-valley g_t is largely cancelled step to step

Momentum is an exponential moving average

Unroll the recursion $v_t = \beta v_{t-1} + g_t$:

$$v_t = g_t + \beta g_{t-1} + \beta^2 g_{t-2} + \beta^3 g_{t-3} + \dots$$

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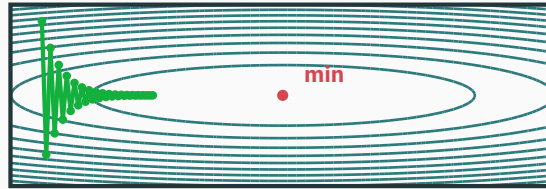
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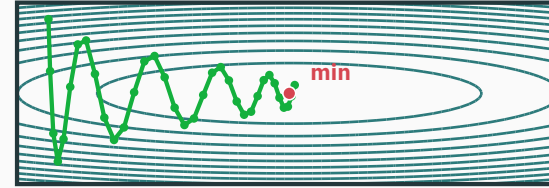
A **geometrically weighted sum** of all past gradients. Recent gradients dominate; old ones fade by β each step. It **remembers the direction**.

Why momentum helps

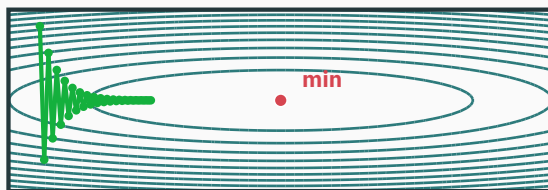
plain GD



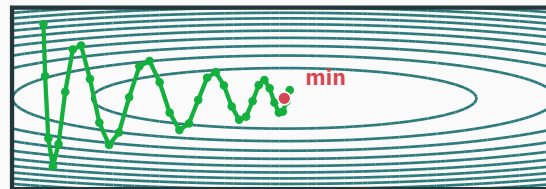
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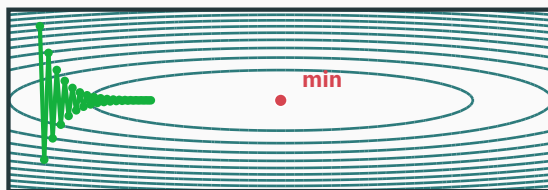


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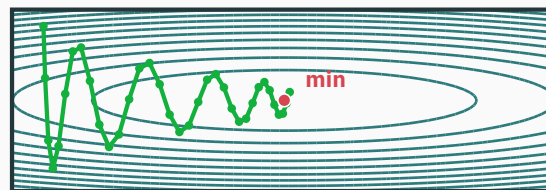


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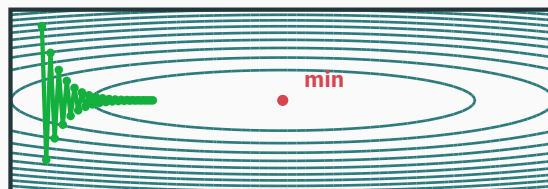


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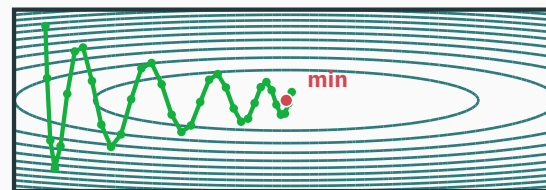


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- **consistent** directions (x) $\rightarrow$ gradients agree $\rightarrow$ they **accumulate** and accelerate;

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- **oscillating** directions (y) $\rightarrow$ gradients flip sign $\rightarrow$ they **cancel** in v_t ;
- **consistent** directions (x) $\rightarrow$ gradients agree $\rightarrow$ they **accumulate** and accelerate;
- averaging also **smooths** stochastic noise.

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$$m_1 = 0.9 \cdot 0 + 0.1 \cdot 10 = 1.0$$

one gradient of 10 only nudges the average to 1.0 — the memory of $m_0 = 0$ still dominates.

Carry $m_1 = 1.0$ forward; $\beta = 0.9$, next gradients 8 then 11.

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raw gradients average ~ 9.7 , but starting from 0 the EMA is still climbing at $m_3 = 2.63$ — early averages are biased **low** (Adam will correct this).

How much does momentum remember?

The weights β^k decay geometrically; the **effective memory** is

$\frac{1}{1 - \beta}$ steps.

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Higher β = longer memory = more smoothing, but slower to react to a real change of direction.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/optimizer-race

Turn momentum on and off in the ravine and sweep β — watch zigzags collapse into a smooth, accelerating glide down the valley.

Adaptive learning rates

One global η is too blunt

Different parameters live on different scales and see different gradient sizes.

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give every coordinate its own effective learning rate

The adaptive form

Scale each coordinate j by the size of **its own** recent gradients:

$$\theta_{t+1,j} = \theta_{t,j} - \frac{\eta}{\sqrt{s_{t,j}} + \varepsilon} g_{t,j}$$

The adaptive form

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- $s_{t,j}$ — a running estimate of the **squared** gradient in coordinate j ;
- ε — a small constant ($\sim 10^{-8}$) to avoid dividing by zero.

RMSProp: an EMA of squared gradients

Keep an exponential average of g^2 , **elementwise**:

$$s_t = \rho s_{t-1} + (1 - \rho) g_t^2$$

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Big recent gradients $\Rightarrow$ large $\sqrt{s_t} \Rightarrow$ **smaller** step. Persistently small gradients $\Rightarrow$ **larger** step. The step self-normalizes.

Why divide by the RMS?

Dividing by $\sqrt{s_t}$ makes the update **scale-invariant** per coordinate:

$$\frac{g_{t,j}}{\sqrt{s_{t,j}}} \approx \pm 1$$

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Steep and shallow directions get **comparable** step sizes — exactly what the ravine needed, without hand-tuning per axis.

Steep axis sees $g = 10$, shallow axis sees $g = 0.1$; take $\rho = 0.9$, $s_0 = 0$:

Worked example: RMSProp on two axes

D DERIVATION

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Divide each gradient by its own $\sqrt{s}$ (ignore ε):

$$\frac{g_{\text{steep}}}{\sqrt{s_{\text{steep}}}} = \frac{10}{\sqrt{10}} \approx 3.16, \quad \frac{g_{\text{shallow}}}{\sqrt{s_{\text{shallow}}}} = \frac{0.1}{\sqrt{0.001}} \approx 3.16$$

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a $100 \times$ gap in raw gradient becomes the same effective step

self-normalization: both axes move at rate $\sim \eta$, so the ravine no longer zigzags.

Adam



Adam = momentum + adaptivity

Keep **two** moving averages of the gradient:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (\text{1st moment — direction})$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (\text{2nd moment — magnitude})$$

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$$m_t = \text{smoothed direction} \cdot v_t = \text{smoothed scale}$$

A subtle bug: initialization bias

We start $m_0 = 0$ and $v_0 = 0$. After one step with constant g :

$$m_1 = (1 - \beta_1)g = 0.1g \quad (\text{way below } g)$$

A subtle bug: initialization bias

We start $m_0 = 0$ and $v_0 = 0$. After one step with constant g :

$$m_1 = (1 - \beta_1)g = 0.1g \quad (\text{way below } g)$$

Early moving averages are biased **toward zero** — worst at $t = 1$, since there is nothing yet to average.

Bias correction

Divide out the shrinkage factor $1 - \beta^t$:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

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as $t \rightarrow \infty$, $\beta^t \rightarrow 0$ and the correction fades away.

Worked example: Adam step 1

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$$m_1 = 0.9 \cdot 0 + 0.1 \cdot 4 = 0.4$$

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2nd moment — EMA of the squared gradient (magnitude):

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both are far **below** the true scale ($g = 4$, $g^2 = 16$) — biased toward 0 by the cold start.

Undo the bias by dividing out $1 - \beta^t$ (here $t = 1$):

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bias correction rescues m_1, v_1 back to the true gradient scale — the step is exactly $-\eta$.

Worked example: Adam step 2

D DERIVATION

$g_2 = 6$ (now $t = 2$), carrying $m_1 = 0.4$, $v_1 = 0.016$ from before:

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$g_2 = 6$ (now $t = 2$), carrying $m_1 = 0.4$, $v_1 = 0.016$ from before: **1st moment** — decay the old average, add 10% of the new gradient:

$$m_2 = 0.9 \cdot 0.4 + 0.1 \cdot 6 = 0.96$$

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the bias factor is now $1 - \beta^2$, weaker than at $t = 1$ — the correction is already fading.

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each coordinate moves by roughly $\pm\eta$, regardless of the raw gradient size.

Adam intuition

$$\theta_{t+1} = \theta_t - \eta \frac{\underbrace{\hat{m}_t}_{\text{which way}}}{\underbrace{(\sqrt{\hat{v}_t} + \varepsilon)}_{\text{how big}}}$$

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- numerator $\hat{m}_t$ — a **momentum-smoothed direction**;
- denominator $\sqrt{\hat{v}_t}$ — a **per-coordinate magnitude** that self-normalizes;
- together: a smoothed, scale-free step in every coordinate.

Adam defaults

Hyperparameter	Default
β_1 (momentum)	0.9
β_2 (RMS)	0.999
ε	10^{-8}
η (learning rate)	10^{-3} (still tune this)

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These defaults “just work” on a huge range of problems — a big reason Adam is the default optimizer.

What extra information does Adam keep beyond plain gradient descent?

A. Only the current loss value

B. A first-moment direction and a second-moment magnitude estimate

C. A full Hessian matrix

D. The complete training dataset

Correct: B — A first and a second moment

Why: The first moment smooths direction like momentum; the second tracks per-coordinate gradient scale for adaptive steps.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/optimizer-race

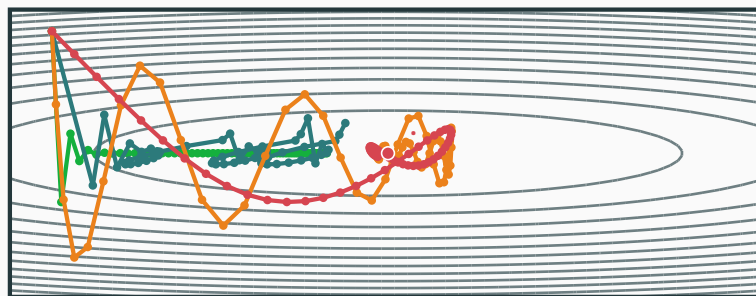
Race GD, SGD, momentum and Adam on the same surface. Watch Adam's per-coordinate scaling walk straight down a ravine that makes plain GD zigzag.

Putting the optimizers side by side

$\mathcal{L} = x^2 + 50y^2$, identical start point:

Same loss, four optimizers

$\mathcal{L} = x^2 + 50y^2$, identical start point:



GD crawls · **SGD** jitters · **momentum** overshoots then settles · **Adam** cuts across

Summary table

Optimizer	Memory	Adaptive?	Strength
GD	none	no	exact gradient, stable
SGD	none	no	cheap, noisy, explores
Momentum	1st moment	no	smooths valleys, accelerates
Adam	1st + 2nd	yes	robust default, per-coord scaling

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- **Either way, the LR schedule matters** as much as the optimizer choice.

Pragmatic rule: start with AdamW + a cosine schedule; reach for SGD+momentum when you are chasing the last bit of test accuracy.

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decoupled decay $\Rightarrow$ the modern default for training large models

Learning-rate schedules

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- **early** training: far from any minimum → want **large** steps to make progress;

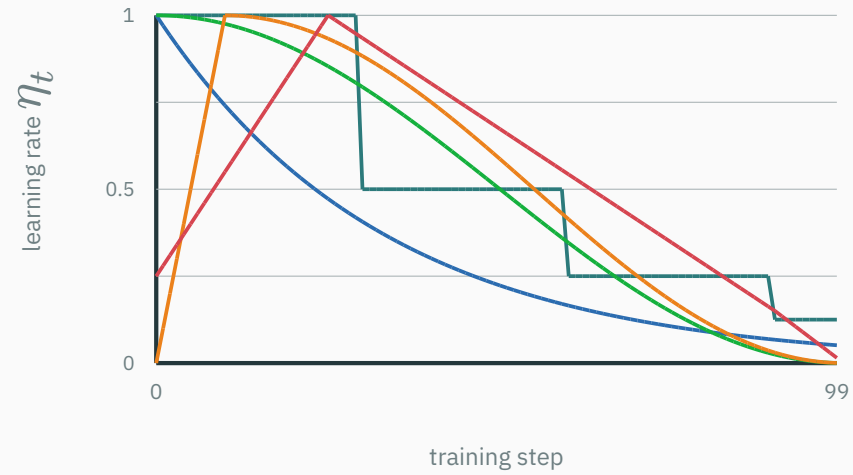
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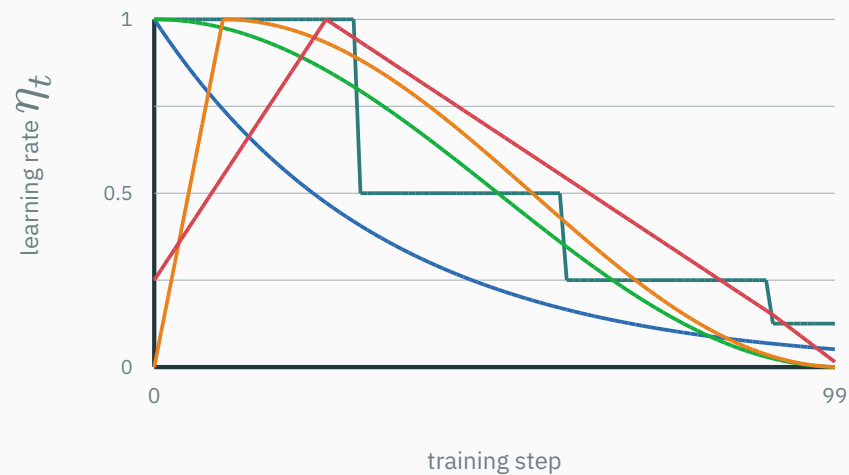
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so decay the learning rate over training





step · exponential · cosine · warmup+cosine · one-cycle

Warmup

Start from **near zero** and ramp up over the first T_{warmup} steps, then decay:

$$\eta_t = \eta_{\max} \cdot \frac{t}{T_{\text{warmup}}} \quad \text{for } t \leq T_{\text{warmup}}$$

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Early gradients are large and unreliable (moments not yet warmed up); a big LR then can **blow up** training. Warmup is near-essential for Transformers.

Cosine decay + warmup — the modern default

$$\eta_t = \frac{1}{2} \eta_{\max} \left(1 + \cos \left(\pi \frac{t - T_{\text{warmup}}}{T - T_{\text{warmup}}} \right) \right)$$

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- smooth decay from $\eta_{\max}$ down to ~ 0 ;
- no sharp drops (unlike step decay);
- pairs naturally with a short linear warmup.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/lr-schedule-visualizer

Compose warmup length, peak LR, and decay shape (step / exponential / cosine / one-cycle) and watch the resulting η_t curve — and how it changes the loss trajectory.

Optimization pathologies

Too much gradient noise

SGD noise helps **explore** — but past a point it **prevents convergence**.

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If $\eta \text{Var}(\varepsilon_t)$ stays large, the iterate **bounces around** the minimum forever.

Remedies: decay η , grow the batch, or use momentum to average the noise out.

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Backprop multiplies Jacobians across layers:

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fixes live in **architecture**: activation choice, init, normalization, residual connections, gradient clipping.

Saddle points

In high dimensions, most points with $\nabla \mathcal{L} = 0$ are **saddles**, not minima — the Hessian has **mixed-sign** eigenvalues.

Saddle points

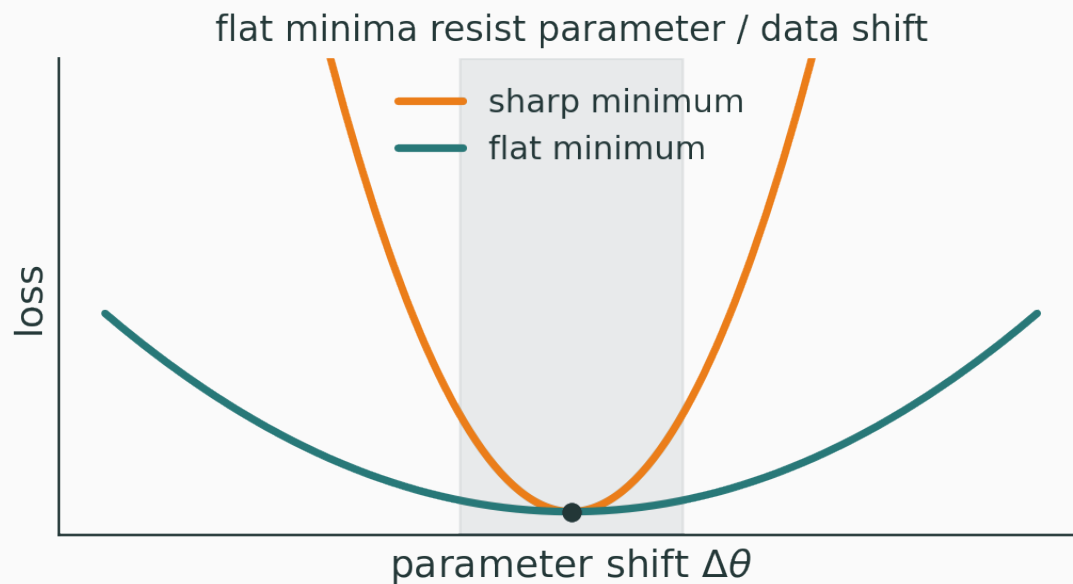
In high dimensions, most points with $\nabla \mathcal{L} = 0$ are **saddles**, not minima — the Hessian has **mixed-sign** eigenvalues.

Plain gradient descent can **stall** on the flat directions of a saddle. SGD's noise is a feature here: it **kicks the iterate off** the saddle and lets training continue.

Sharp vs flat minima

V VISUAL

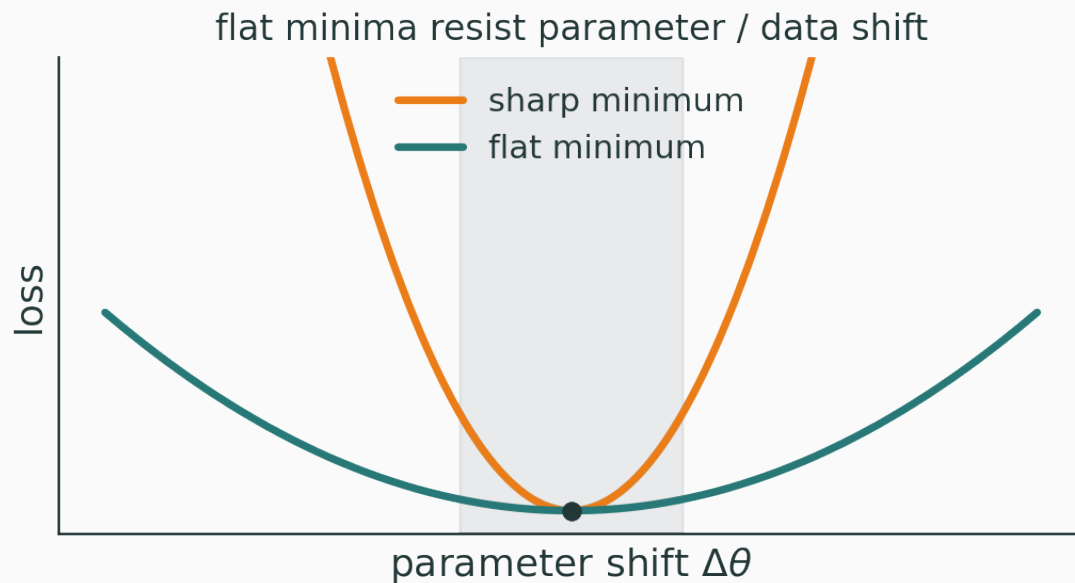
★ OPTIONAL



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Flat minima are widely associated with **better generalization**: a small shift in parameters or data barely changes the loss. Small-batch SGD tends to prefer flatter minima. (an active, subtle research area)

The training loop, end to end

One optimization step in PyTorch

I INTERACTIVE

```
for x, y in loader:           # minibatch
    optimizer.zero_grad()      # clear old gradients
    logits = model(x)          # forward pass
    loss = loss_fn(logits, y)  # scalar loss
    loss.backward()             # backprop -> param.grad
    optimizer.step()            # update rule uses grads
```


One optimization step in PyTorch

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forward → loss → **backprop (gradients)** → **optimizer (update)** — every lecture so far, in six lines.

The step, three ways

```
# SGD
p -= lr * p.grad

# SGD + momentum
v = beta * v + p.grad
p -= lr * v

# Adam (per parameter)
m = b1*m + (1-b1)*p.grad
v = b2*v + (1-b2)*p.grad**2
mhat = m/(1-b1**t); vhat = v/(1-b2**t)
p -= lr * mhat / (vhat.sqrt() + eps)
```

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```

same loop — swap only what `optimizer.step()` does.

Choosing an optimizer in one line

```
opt = torch.optim.SGD(model.parameters(), lr=0.1, momentum=0.9)
opt = torch.optim.AdamW(model.parameters(), lr=3e-4)

sched = torch.optim.lr_scheduler.CosineAnnealingLR(opt, T_max=epochs)
```

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```

The optimizer and the schedule are **two knobs**; both matter. Defaults above are sane starting points.

One-slide summary

Method	Core idea
GD	full-dataset gradient — exact but slow
SGD	sample a batch — unbiased, noisy, cheap
Momentum	average gradients — smooth, accelerate
RMSProp	divide by RMS gradient — per-coord scale
Adam	momentum + adaptive scale + bias correction
LR schedule	big early, small late (warmup + decay)

Final mental model — one idea

Backprop answers: **what direction?**

Optimization answers:

how far, how smoothly, how adaptively?

$\text{gradient} = \text{direction} \cdot \text{optimizer} = \text{the art of the step}$

Backprop gives the gradient; the optimizer turns it into learning.

Next: **regularization and generalization** — why the network that fits the training set also works on data it has never seen.